

Music-Gen v3 SPINE Milestone — Cycles 14-16

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Music-Gen v3 SPINE Milestone — Cycles 14-16

Abstract

This report covers three consecutive heartbeat cycles of the M-V3-SPINE milestone on the reference track Chicken Grease (source SHA-16 31a164f845f8e27e). All three cycles run under the wait-on-operator cadence policy landed in Cycle 8, and neither of that policy’s two break-glass triggers — a live-guidance operator directive or an auditor CRITICAL finding — fires at any point in the arc. The three cycles land the policy’s prescribed four-slot shape three more times, extending the campaign’s consecutive-heartbeat streak from five (through Cycle 13) to eight and its clean-audit streak from nine (through Cycle 13) to twelve. Every earlier verdict remains byte-identical on disk; the anchor snapshot grows from 156 anchors at Cycle 13 through 166, 176, and 186 across this arc at a steady +10 per cycle; the three-way rubric-v2 integrity chain, the ten-cycle venv byte-identity, the append-only verdict chain, and the Mode-2 lock on the torch-2.13 reproduction all remain intact. The one minor bookkeeping variance that had persisted across the previous heartbeat arc — a small drift between the worker’s declared promise_check warning baseline and the auditor’s live count — resolves cleanly at Cycle 16 with a zero delta. M-V3-SPINE-1 remains gated on operator ear per Fixed Decision 6; all downstream milestones remain frozen.

1. Continuity from Cycles 11-13

The Cycles 11-13 arc had established that the heartbeat cadence could hold indefinitely under continued operator absence without either drift on any load-bearing anchor or the accumulation of false progress. The three cycles reported here — Cycles 14, 15, and 16 (the sixth, seventh, and eighth consecutive heartbeats in the campaign) — do not change that story. They demonstrate it three more times, close one of the two remaining minor observations from the earlier arc, and continue to accumulate audit trail at a stable rate.

2. Methodology carried across all three cycles

The methodology is unchanged from Cycles 11-13. Every new script sets the pinned environment (PYTHONHASHSEED=0, SOURCE_DATE_EPOCH=1756463424, TZ=UTC, LC_ALL=C.UTF-8, and single-threaded BLAS via OMP/MKL/OPENBLAS_NUM_THREADS=1); every top-level invocation uses /usr/bin/python3 under an AST-verified interpreter guard; every new file is generated deterministically; every new script is AST-scanned to fail on any network or PRNG import.

The three integrity chains established earlier all continue to hold: the three-way rubric-v2 chain (document c49db5a12e955f26...451a == pinned hash file == each cycle's verdict.rubric_hash_v2), the cadence-policy hash chain (0be540365c8c03ad...c7f2 == its pinned hash file), and the Cycle 7 torch-probe module SHA (b54adadd...9af3d, byte-identical across the entire campaign to date). The backref pattern established at Cycle 10 continues: each cycle's verdict pins the previous cycle's verdict.json SHA in c<N-1>_backref_sha, resolved from disk at emit time and re-verified by the auditor's walker test.

3. Heartbeat shape (identical across Cycles 14, 15, 16)

Each of the three cycles delivers the four artifacts, four housekeeping ledger rows, and twelve-case test suite specified by the cadence policy.

3.1 Track 1 — Torch-213 dry-run liveness roll-forward

A per-cycle caller (scripts/v3_spine/torch213_reproduce_probe_c<N>.py) imports the read-only Cycle 7 probe module and runs Mode 1 only. Four checks are asserted against a cumulative baseline that extends by one cycle each pass:

Cycle	Baseline chain	attribution_verdict
14	vs c7+c8+c9+c10+c11+c12+c13	ENV_DRIFT_PROBE_CANDIDATE_ FOUND_C14_DRY_RUN_ROLL_FORWARD
15	vs c7..c14	ENV_DRIFT_PROBE_CANDIDATE_ FOUND_C15_DRY_RUN_ROLL_FORWARD
16	vs c7..c15	ENV_DRIFT_PROBE_CANDIDATE_ FOUND_C16_DRY_RUN_ROLL_FORWARD

The four asserted properties — torch.__version__ = 2.13.0+cpu, torch.__file__ = /usr/local/lib/python3.11/dist-packages/torch/__init__.py, drafted reproduction command (binary form and -m muscriptor.cli module form) byte-identical to the Cycle 7 pin, and venv (workspace/learned_transcribers_venv/) directory-manifest SHA a86205175728d58f0a96ad02fc1ab1ac9e35f06c5ed568a960ed1ff261f83a74 — pass at every cycle. network_syscall_attempted=false and venv_unchanged=true at every cycle. Mode 2 remains awaiting_operator_green_light throughout the arc; the durable Cycle 7 lock has now held across the ten cycles from Cycle 7 through Cycle 16.

The inner checks_vs_baseline key that carries the growing baseline chain (venv_manifest_matches_c7_c8_c9_c10_c11_c12_c13_c14_c15 at Cycle 16, for example) formalizes the naming convention foreshadowed by the Cycle 14 auditor and cleanly extends by one segment each cycle.

3.2 Track 2 — Anchor preservation

At the start of each cycle every read-only artifact is snapshotted; at end of cycle every anchor is re-hashed and byte equality asserted. The list grows by exactly ten anchors per cycle as the

previous cycle’s own artifacts (verdict, torch-probe JSON, three anchor-preservation JSONs, three driver scripts, twelve-case test file, cycle-scoped report) join the list.

Cycle	Anchor count	all_match	n_diff	Target
14	166	true	0	≥ 165
15	176	true	0	≥ 175
16	186	true	0	≥ 185

Every locked script remained byte-identical across all three cycles, including the c7 probe module b54adadd...9af3d (now a ten-cycle chain), the Cycle 12 probe module 71da91b3...03bc1 (added to the locked list at Cycle 12), and every earlier delivered verdict.

3.3 Track 3 — Verdict emission

Each cycle emits its verdict at data/v3/deliveries/31a164f845f8e27e/cycle<N>/verdict.json. The three verdict SHAs on disk at the end of this arc:

Cycle	Verdict	On-disk SHA	cycles_since_last_operator_ input
14	V3_SPINE_C14_ HEARTBEAT_pending_ operator	pinned by c15 backref	10
15	V3_SPINE_C15_ HEARTBEAT_pending_ operator	a88672b35b59e754...f9712	11
16	V3_SPINE_C16_ HEARTBEAT_pending_ operator	d251c51bb00e3665c694997c136ac0eea7668824865a43e13a28bcd736deddd8	12

Each verdict carries blocked_on_operator=true, cadence_mode=heartbeat, a prior_cycles list enumerating every earlier cycle in the arc (Cycle 16’s list has twelve entries, c4 through c15), the previous cycle’s c<N-1>_backref_sha resolved on disk at emit time, and the two operator-facing WAV pins with operator_ear_pending_fd6 status (Method A cc919559b4508b6b...; Method B f40796be982998b0...). The three-way rubric-v2 chain is byte-equal at each emission.

3.4 Track 4 — Housekeeping

Each cycle emits eight ledger events in strict order under a single cycle-scoped run_id (run-2026-09-02T<HH>0000Z, HH = 18/19/20 for Cycles 14/15/16 respectively):

1. M-V3-SPINE-1/torch213-reproduce-probe-c<N>-completed
2. M-V3-SPINE-1/anchor-preservation-pre-c<N>-verified
3. M-V3-SPINE-1/anchor-preservation-post-c<N>-verified
4. M-V3-SPINE-1/verdict-c<N>-emitted (status=action_required)
5. M-INGEST-1/egress-probe-cycle<N>
6. _plan/register-c<N>-v3-spine-sub-leaves
7. _infra/adopt-cycle<N>-tests
8. _archive/cycle-<N>-scratch (timestamped ts+1s after the physical move)

The egress probe row continues to record HTTP 429 + tv_embedded on every attempt. The promise_check gate returned zero errors at every cycle.

3.5 Test discipline

Each cycle adds `tests/test_v3_spine_c<N>.py` with the same twelve-case shape used since Cycle 9, and re-runs prior heartbeat suites and the generic verdict-walker invariant test. Cumulative sanity-floor totals through this arc:

Cycle	New cases	Regression suites re-run	Total green
14	12	c13, c12, c11, c10, c9, generic	68/68
15	12	c14 through c9 + generic	80/80
16	12	c15 through c9 + generic	92/92

The auditor independently re-ran the newest cycle’s suite and the generic invariant suite at each cycle under the mandated environment; all runs passed live.

4. Cycle-specific notes

The three cycles are structurally identical; the differences are cosmetic or bookkeeping.

Cycle 14 — sixth consecutive heartbeat. Anchor list grows to 166 (+10). No new observations of note.

Cycle 15 — seventh consecutive heartbeat. Anchor list grows to 176 (+10). The Cycle 15 audit issued zero critical, zero moderate, and one minor observation (MINOR-1): the worker’s declared `promise_check` warning baseline of 2768 was 2770 on the auditor’s live re-run, a +2 delta consistent with the pattern that had been drifting +1/+2 through Cycles 12–15 as new heartbeat-cycle test files landed without full per-file sub-leaf adoption. The observation was cosmetic — all content-level assertions passed — and the fix was queued into the Cycle 16 sub-leaf accounting envelope.

Cycle 16 — eighth consecutive heartbeat. Anchor list grows to 186 (+10). The Cycle 15 minor observation resolves cleanly: the worker declared 2774 warnings, and the auditor’s live re-run returned exactly 2774 (zero delta). The `_infra/adopt-cycle16-tests` sub-leaf carried its own artifact list correctly, and the `data/v3_spine/cycle16/` data directory sub-leaf was registered with its expected envelope, closing the Cycles 12–15 heartbeat-churn accounting pattern. Cycle 16 was 0 CRITICAL / 0 MODERATE / 0 MINOR — the first fully-clean audit of the heartbeat arc.

5. Cumulative state at end of arc

M-V3-SPINE-1 has now sat in `blocked_on_operator` state for twelve cycles since Cycle 5 without regression on any load-bearing invariant. Twelve `LANDS_pending_operator` verdicts sit on disk (Cycles 4 and 5 as substantive-cycle landings, Cycle 6 as the two-track close-out, Cycle 7 as the last three-track cycle, Cycle 8 as the moderate-fix cycle, and Cycles 9 through 16 as the eight heartbeats), all byte-identical to their emission-time hashes. Both operator-facing A/B candidates remain unchanged: Method A at `data/v3/deliveries/31a164f845f8e27e/operator_section/full_reconstruction_operator_section.wav` (SHA `cc919559b4508b6b...`) and Method B at `data/v3_spine/rc7_v2_v3_paths/rc7_v2_v3_paths_full_reconstruction.wav` (SHA `f40796be982998b0...`).

The three operator-facing decisions queued at the end of Cycles 4–6 remain queued:

1. **Ear verdict on either A/B pair.** A positive verdict opens M-V3-FOCUS-1 (Chicken Grease anchor plus four SHA-256-tiebreak picks from `data/recreate_v2/focus_set_v2.json`).
2. **Approval to execute the torch-2.13 reproduction (Mode 2).** The drafted commands remain pinned byte-identically across Cycles 7 through 16.

3. Canonicity choice between the two mix chains.

Downstream milestones M-V3-FOCUS-1, M-V3-CORPUS-1, M-V3-RULES-1, M-V3-EAR-1, and M-V3-GEN-1 remain frozen.

Three quantitative observations describe the heartbeat regime after this arc:

- **Twelve consecutive clean audit cycles (Cycles 5-16)** with roughly a hundred cumulative live SHA spot-checks resolved on disk and zero fabrications detected.
- **Ten-cycle byte-identical venv manifest** (Cycles 7-16), demonstrating that the c3-era torch 2.13.0+cpu attribution candidate remains reproducible on demand in Mode 1 without any environment drift accumulating during the wait.
- **Steady-state +10 anchors per cycle** growth confirmed across five heartbeat cycles at this cadence, projecting to about 196 anchors at a hypothetical Cycle 17 heartbeat.

6. Conclusions

The Cycles 14-16 arc is not a story of change; it is a story of the campaign's steady-state mode holding without drift. Eight consecutive heartbeats have now landed under the Cycle 8 policy, three of them in this arc, and the twelve-cycle clean-audit streak proves that the discipline invariants encoded in the policy — rubric hash chain, cadence policy hash chain, venv byte-identity, anchor preservation, append-only verdicts, Mode-2 lock, four housekeeping rows, twelve-case test suite — are automatically enforced at essentially zero engineering cost. The single observed minor variance persisting from earlier heartbeat cycles — the `promise_check` warning-count bookkeeping delta — closes cleanly at Cycle 16 with a zero delta and a first fully-clean audit of the heartbeat era.

The unresolved work remains not code but operator judgment. Until any one of the three queued operator decisions arrives in a subsequent cycle's `live_guidance`, the heartbeat cadence continues, the audit trail continues to accrete, and the milestone remains available for immediate positive resolution as soon as an ear verdict lands.

Appendix: Implementation Details

A.1 Delivered artifacts by cycle

Cycle 14, verdict at `data/v3/deliveries/31a164f845f8e27e/cycle14/verdict.json`; working artifacts at `data/v3_spine/cycle14/torch213_reproduce_probe_c14.json` and `anchor_preservation_{pre,post,}_c14.json`.

Cycle 15, verdict at `data/v3/deliveries/31a164f845f8e27e/cycle15/verdict.json` (SHA `a88672b35b59e754...f9712`); working artifacts at `data/v3_spine/cycle15/`.

Cycle 16, verdict at `data/v3/deliveries/31a164f845f8e27e/cycle16/verdict.json` (SHA `d251c51bb00e3665c694997c136a`); torch probe JSON at `data/v3_spine/cycle16/torch213_reproduce_probe_c16.json` (SHA `95ab-cad4b8e93f93e036311575bcd3736207c6dce7ea549de7313b33d43e6f9`); anchor diff at `data/v3_spine/cycle16/anchor_preservation_c16.json` (SHA `80f34565cc285e4e6b2f6c050e235389e7903fabd8defccb5425f486858e76eb`).

Cycle-scoped reports at `docs/v3_spine_report_cycle14.md`, `docs/v3_spine_report_cycle15.md`, `docs/v3_spine_report_cycle16.md`.

A.2 Scripts

Per cycle: `scripts/v3_spine/torch213_reproduce_probe_c<N>.py`, `scripts/v3_spine/anchor_preservation_c<N>.py`, `scripts/v3_spine/verdict_c<N>.py`. One-shot ledger emitters born under `tools/stale/cycle<N>_v3_spine_scratch/` per cycle.

Cycle 7 probe module locked at b54adadd1f4f04028d4d931710cc02b722c82eb027728a97adfa8b20c9af3d (ten-cycle byte-identical chain through Cycle 16). Cycle 12 probe module locked at 71da91b3e4755e8ad881fa561884ce136aa61d6526bee2ba4fa5f98ef3e03bc1. Cycle 15 probe module SHA 65aaf0e6ea1919c7c796c8928b347283de5792e17a93459d5b8e87ebe7fbec62. All other locked scripts (render_stem.py, mix_match_operator_section.py, rc7_v2_rerun.py, rc7_v2_rerun_v3_paths.py, rc7_mix_balance.py) byte-identical across the arc.

A.3 Tests

tests/test_v3_spine_c14.py, tests/test_v3_spine_c15.py, tests/test_v3_spine_c16.py — twelve cases each; tests/test_verdict_sha_fields_resolve_on_disk.py — eight generic-invariant cases. Full sanity-floor regression re-run at every cycle. Cycles 14/15/16 tallies: 68/68, 80/80, 92/92 green.

A.4 Integrity chains

Three-way rubric-v2 chain: docs/v3_spine_rubric_v2.md SHA c49db5a12e955f26c001165ad6e8f9d191bc26bfd96e24c1b1 == data/v3_spine/rubric_hash_v2.txt content == each cycle's verdict.rubric_hash_v2. Cadence policy chain: docs/wait_on_operator_cadence_policy.md SHA 0be540365c8c03ad38a15478fbad0fe32bf5ea4118e33ef3eed == its pinned hash file.

Venv directory-manifest byte-identity across ten cycles (Cycles 7 through 16): a86205175728d58f0a96ad02fc1ab1a

Verdict backref chain: Cycle 15 pins Cycle 14's verdict SHA; Cycle 16 pins Cycle 15's a88672b35b59e754...f9712; auditor re-verifies at each pass.

A.5 Ledger schedule

Each cycle emits eight events under run_id = run-2026-09-02T<HH>0000Z (HH = 18/19/20), timestamps T<HH>:00:00Z for the first seven events and T<HH>:00:01Z for the post-move archive row.

A.6 Environment pins

PYTHONHASHSEED=0; SOURCE_DATE_EPOCH=1756463424; TZ=UTC; LC_ALL=C.UTF-8; OMP_NUM_THREADS=1, MKL_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1; interpreter /usr/bin/python3; mido==1.3.3; SoundFont SHA 74594e8f...1cb0; MuScriptor model SHA ac80adbd...7fb97ec.

A.7 Anchor growth profile

Cycle 13 → Cycle 14: 156 → 166 (+10); Cycle 14 → Cycle 15: 166 → 176 (+10); Cycle 15 → Cycle 16: 176 → 186 (+10). Growth per cycle corresponds to the ten heartbeat-cycle-specific artifacts that lock at cycle close (verdict, torch-probe JSON, three anchor-preservation JSONs, three driver scripts, twelve-case test file, cycle-scoped report document).

A.8 Cumulative audit statistics

Twelve consecutive clean audit cycles (Cycles 5–16); roughly one hundred cumulative live SHA spot-checks across the arc with zero fabrications detected; 0 CRITICAL and 0 MODERATE findings surfaced anywhere in the eight-cycle heartbeat era; the one minor observation persisting from Cycles 12–15 (promise_check warning-count bookkeeping delta) resolved cleanly at Cycle 16 with zero delta.

A.9 Source sessions

Cycle	Researcher	Worker	Auditor
14	be26d81d-2dce-4ef1-8725-95591e62185f	9ffb70b1-ca94-4e12-b359-3860f1ab757b	9af1a867-6d9e-41af-ba79-5d39b2f8510d
15	d544ed44-3491-47a0-b24c-92c837f2b56b	d15bb16e-5863-4f15-8a37-f63678a61caf	a7306871-afe4-4c18-aaaf-0a43fc2ed81b
16	8489c552-16a2-4405-9c84-61bb55e1ee7b	70a61995-6ce2-4ef8-bccf-a8dacece25b3	2959de07-d775-428a-ac4e-929d9047ae31